

An Exact Chern-Phase Atlas for the QWZ Thouless Pump

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Abstract

For a two-band pump on the Bloch torus we prove the exact spectrum, direct gap, all lower-band Chern phases, and every gap-closing Dirac jump. The transported charge statement is made only in the filled-band adiabatic limit. Independent symbolic, degree, and lattice-gauge receipts audit the convention-sensitive signs. This revision is the **exact Bloch-gap owner**.

1 Model and complete gap atlas

Orient $\mathbb{T}^2$ by $dk \wedge d\tau$ and define

$$H_m(k, \tau) = d_m(k, \tau) \cdot \sigma, \quad d_m = (\sin k, \sin \tau, m + \cos k + \cos \tau). \quad (1)$$

Here $m \in \mathbb{R}$ and both angular variables have period 2π . Whenever d_m is nonzero, let $n = d_m/|d_m|$ and $P_- = (I - n \cdot \sigma)/2$.

Theorem 1 (Spectrum, gap, and lower-band phase). *The fibre spectrum is $\{-|d_m|, +|d_m|\}$. The family is gapped exactly for $m \notin \{-2, 0, 2\}$, with direct gap*

$$G(m) = 2 \min\{|m+2|, |m|, |m-2|\}.$$

On a gapped chamber P_- is a smooth rank-one complex line bundle. With

$$c_1(P_-) = \frac{1}{2\pi i} \int_{\mathbb{T}^2} \text{Tr}(P_- [\partial_k P_-, \partial_\tau P_-]) dk d\tau, \quad (2)$$

its Chern number is

$$c_1 = \begin{cases} 0, & m < -2, \\ -1, & -2 < m < 0, \\ +1, & 0 < m < 2, \\ 0, & m > 2. \end{cases}$$

Proof of the spectral and gap assertions. Pauli multiplication gives $H_m^2 = |d_m|^2 I$. With $x = \cos k$ and $y = \cos \tau$,

$$|d_m|^2 = m^2 + 2 + 2m(x + y) + 2xy. \quad (3)$$

The right side is affine in each variable separately, so its minimum on $[-1, 1]^2$ occurs at a corner. The four corner values are $(m+2)^2, m^2, m^2, (m-2)^2$. This proves both the closing locus and the stated gap. Smooth functional calculus then gives the displayed lower projector. The Chern computation is proved below in revision one. $\square$

2 Projector sign, degree, and Dirac walls

This revision is the **analytic Dirac-degree owner**. Direct Pauli algebra fixes the convention-sensitive sign:

$$\text{Tr}(P_- [\partial_k P_-, \partial_\tau P_-]) = -\frac{i}{2} n \cdot (\partial_k n \times \partial_\tau n). \quad (4)$$

Consequently (2) is minus the Brouwer degree of $n : \mathbb{T}^2 \rightarrow S^2$. For completeness, a direct differentiation before normalization gives

$$d_m \cdot (\partial_k d_m \times \partial_\tau d_m) = \cos k + \cos \tau + m \cos k \cos \tau. \quad (5)$$

Choose the north pole as regular value away from the walls. Its possible preimages are the four points with $\sin k = \sin \tau = 0$. A point contributes exactly when its mass $M = m + \cos k_0 + \cos \tau_0$ is positive, and its local degree is $\chi(k_0, \tau_0) = \cos k_0 \cos \tau_0$. Thus $\deg(n) = \sum \chi \mathbf{1}_{\{M>0\}}$. Substituting $\mathbf{1}_{\{M>0\}} = (1 + \operatorname{sgn} M)/2$ and using $\sum \chi = 0$ yields

$$c_1 = -\frac{1}{2}[\operatorname{sgn}(m+2) - 2\operatorname{sgn}(m) + \operatorname{sgn}(m-2)]. \quad (6)$$

This proves the remaining part of Theorem 1 without numerical integration. As m increases, the complete wall ledger is

mass	zero (k_0, τ_0)	χ	jump of c_1
-2	$(0, 0)$	+1	-1
0	$(\pi, 0)$	-1	+1
0	$(0, \pi)$	-1	+1
2	(π, π)	+1	-1

The simultaneous pair at $m = 0$ is essential: its total jump is two.